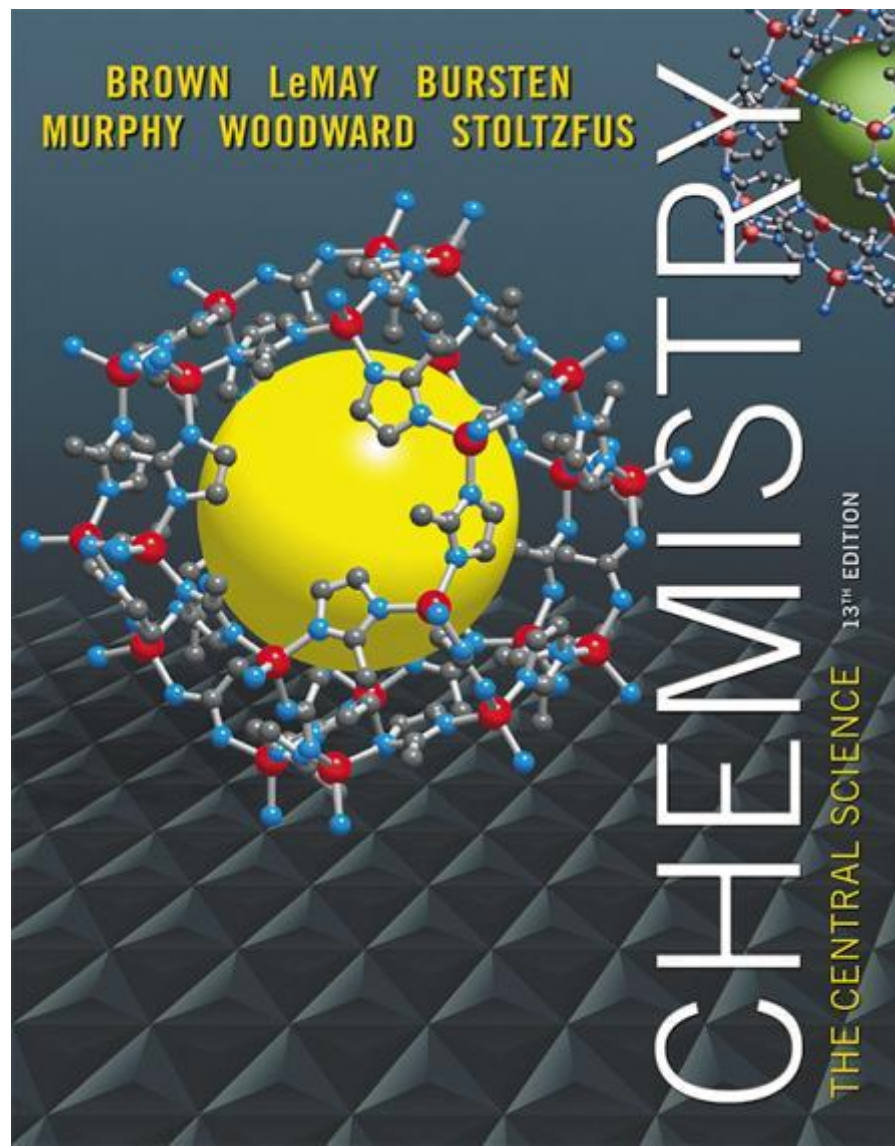


Chapter 2. Molecular Geometries



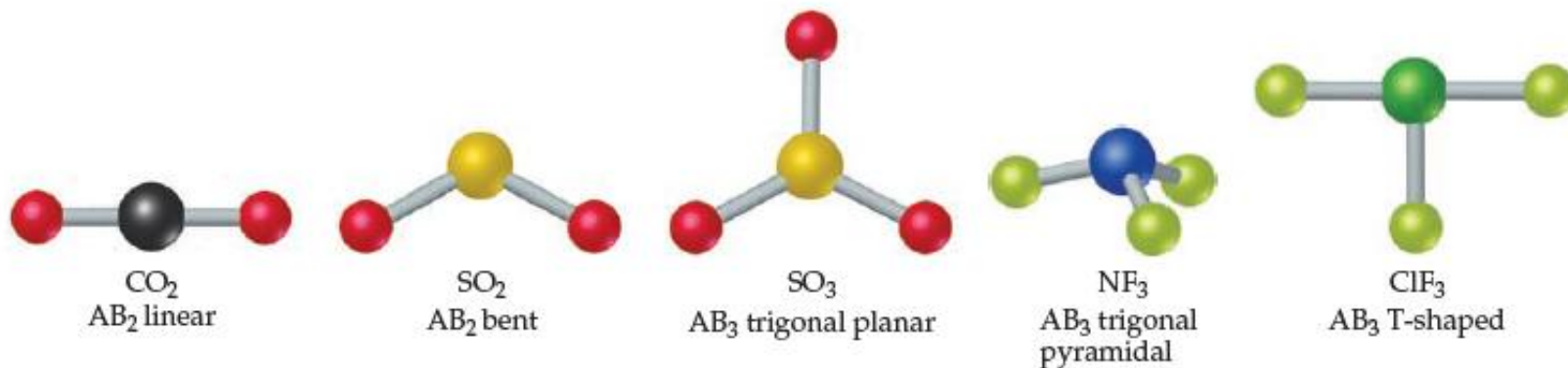
Reference:

Chapter 9

**Molecular Geometry and Bonding
Theories**

Molecular Shapes

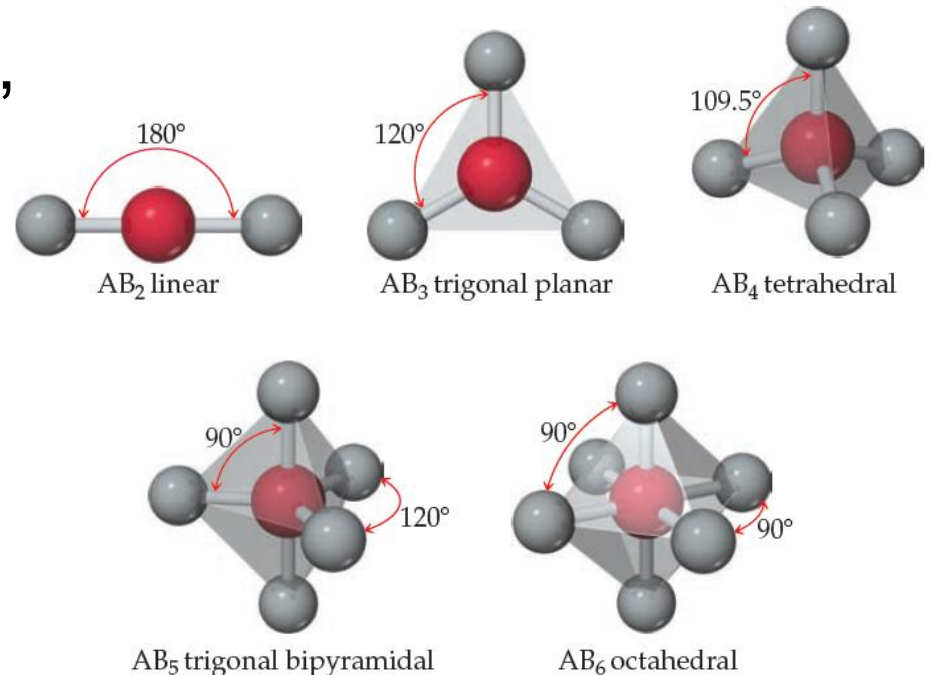
- Lewis Structures show bonding and lone pairs, but do *not* ^{表示} denote shape.
- However, we *use* Lewis Structures to help us determine shapes.
- Here we see some common shapes for molecules with two or three atoms connected to a central atom.



What Determines the Shape of a Molecule?

- Simply put, electron pairs, whether they be bonding or nonbonding, 排斥 repel each other.

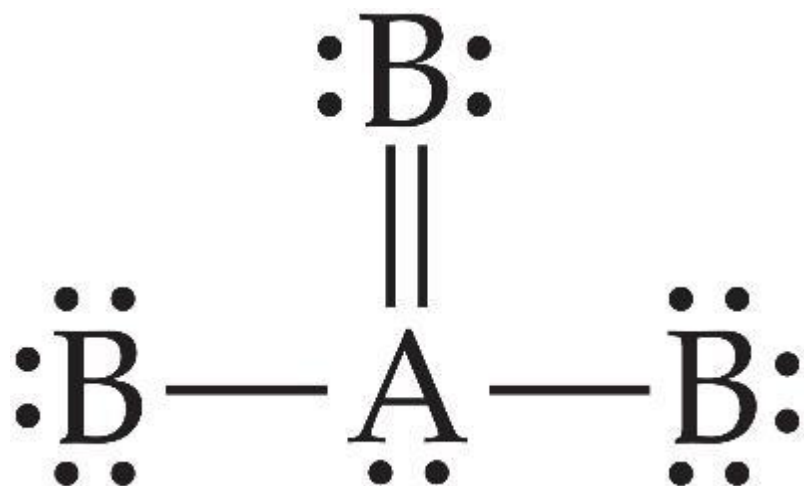
- By assuming the electron pairs are placed as far as possible from each other, we can predict the shape of the molecule.



▲ Figure 9.3 Shapes allowing maximum distances between B atoms in AB_n molecules.

- This is the **Valence-Shell Electron-Pair Repulsion (VSEPR)** model.

Electron Domains



- 1 single/double/triple bond = 1 electron domain
- 1 lone pair = 1 electron domain

- We can refer to the directions to which electrons point as **electron domains**. This is true whether there is one or more electron pairs pointing in that direction.
- The central atom in this molecule, A, has four electron domains.

Electron Domains (电子域) 指的是中心原子周围的电子密度区域, 在价层电子对互斥理论 (VSEPR) 中用于预测分子几何构型。

主要类型:

成键域: 单键、双键或三键各计为一个电子域

孤对域: 未参与成键的电子对

重要性:

电子域相互排斥, 倾向于在空间中尽量远离, 以最小化排斥力, 从而决定分子的基本几何形状。

常见例子:

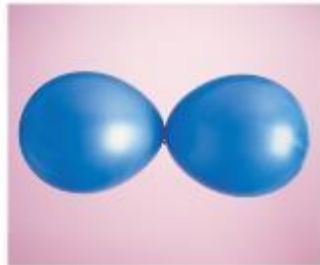
2 个电子域 → 线性 (180°)

3 个电子域 → 三角平面 (120°)

4 个电子域 → 四面体 (109.5°)

孤对电子的排斥力比成键电子对更强, 会压缩键角, 导致实际分子形状与理想几何构型有所不同。

Valence-Shell Electron-Pair Repulsion (VSEPR) Model



Two balloons
linear orientation



Three balloons
trigonal planar orientation



Four balloons
tetrahedral orientation

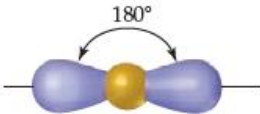
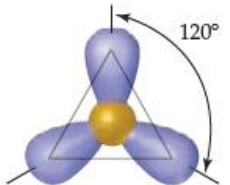
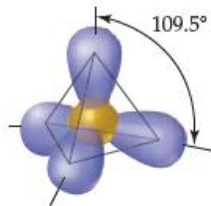
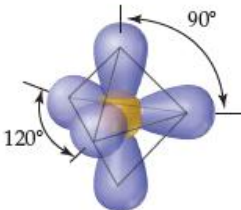
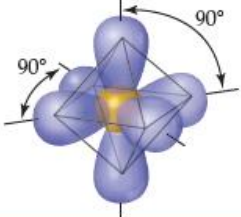
最小化排斥

“The best arrangement of a given number of electron domains is the one that minimizes the repulsions among them.”

(The balloon analogy in the figure to the left demonstrates the maximum distances, which minimize repulsions.)

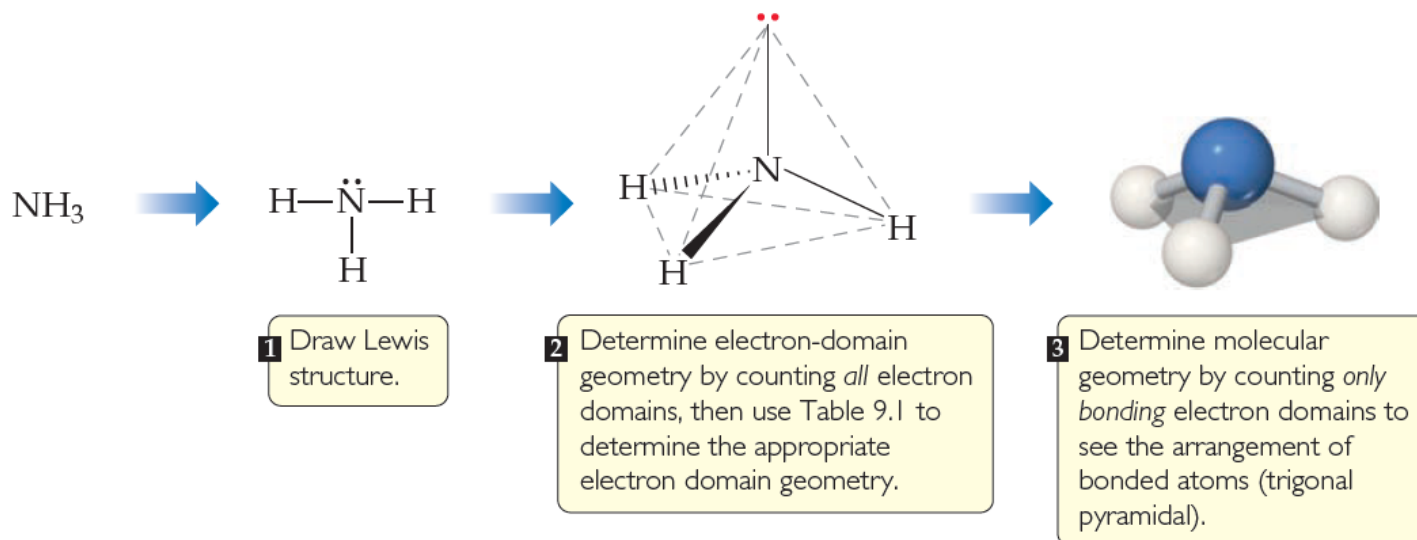
Electron-Domain Geometries

Table 9.1 Electron-Domain Geometries as a Function of Number of Electron Domains

Number of Electron Domains	Arrangement of Electron Domains	Electron-Domain Geometry	Predicted Bond Angles
2		Linear	180°
3		Trigonal planar	120°
4		Tetrahedral	109.5°
5		Trigonal bipyramidal	120° 90°
6		Octahedral	90°

- The Table shows the electron-domain geometries for two through six electron domains around a central atom.
- To determine the electron-domain geometry, count the total number of lone pairs, single, double, and triple bonds on the central atom.

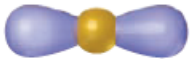
Molecular Geometries



- Once you have determined the electron-domain geometry, use the arrangement of the bonded atoms to determine the **molecular geometry**.
- Tables 9.2 and 9.3 show the potential molecular geometries. We will look at each electron domain to see what molecular geometries are possible.

Linear Electron Domain

Table 9.2 Electron-Domain and Molecular Geometries for Two, Three, and Four Electron Domains around a Central Atom

Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
2	 Linear	2	0	 Linear	$\ddot{\text{O}}=\text{C}=\ddot{\text{O}}$

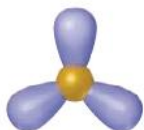
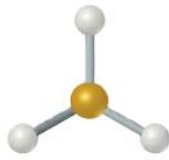
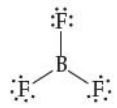
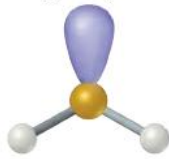
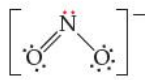
- In the linear domain, there is only one molecular geometry: linear.
- NOTE: If there are only two atoms in the molecule, the molecule will be linear no matter what the electron domain is. (diatomic molecule is linear)

双原子分子

Trigonal Planar Electron Domain

三角平面电子域构型

Table 9.2 Electron-Domain and Molecular Geometries for Two, Three, and Four Electron Domains around a Central Atom

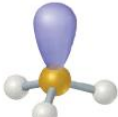
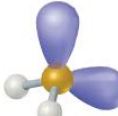
Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
3	 Trigonal planar	3	0	 Trigonal planar	
		2	1	 Bent	

- There are two molecular geometries:
 - trigonal planar, if all electron domains are bonding, and
 - bent, if one of the domains is a nonbonding pair.

Tetrahedral Electron Domain

四面体电子结构域

Table 9.2 Electron-Domain and Molecular Geometries for Two, Three, and Four Electron Domains around a Central Atom

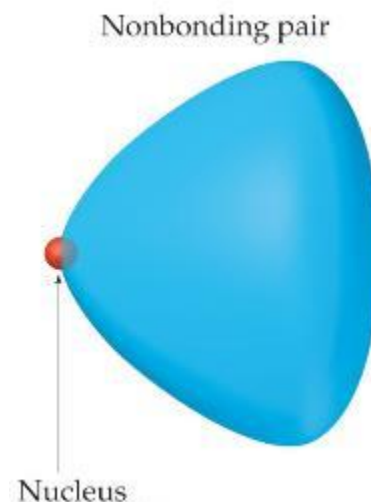
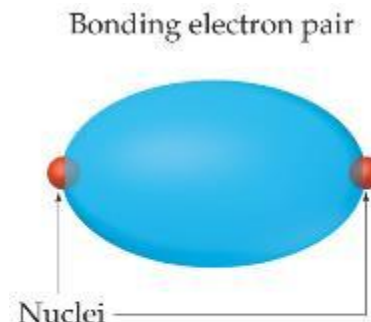
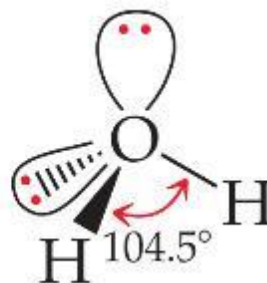
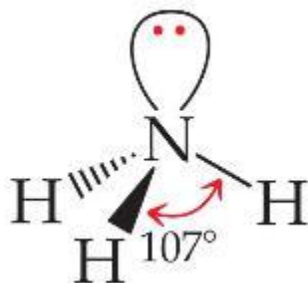
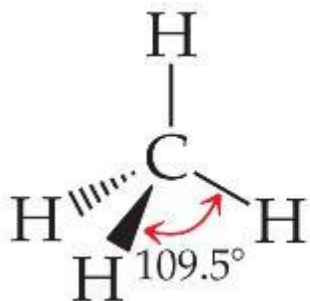
Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
4	 Tetrahedral	4	0	 Tetrahedral	
		3	1	 Trigonal pyramidal	
		2	2	 Bent	

- There are three molecular geometries:
 - tetrahedral, if all are bonding pairs,
 - trigonal pyramidal, if one is a nonbonding pair, and 三角锥形
 - bent, if there are two nonbonding pairs.

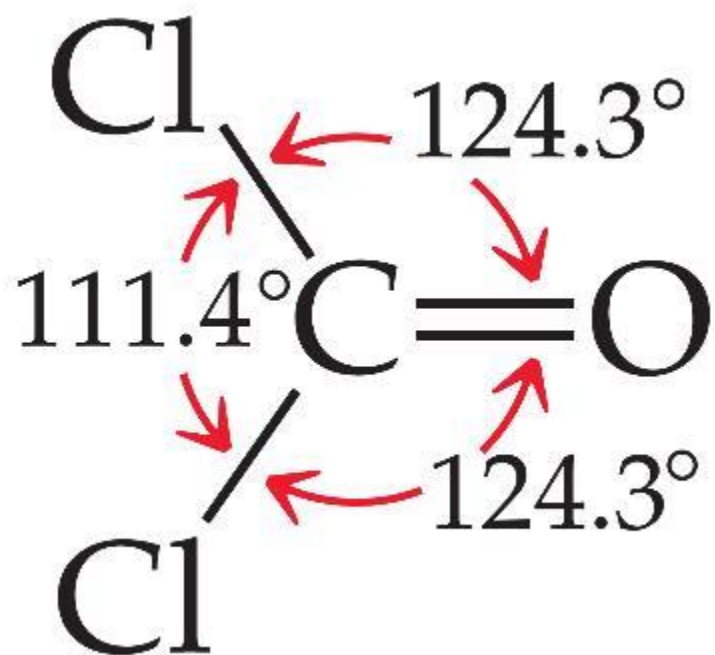
Nonbonding Pairs and Bond Angle

孤电子对和键角

- Nonbonding pairs are physically larger than bonding pairs.
- Therefore, their repulsions are greater; this tends to compress bond angles.



Multiple Bonds and Bond Angles



- Double and triple bonds have larger electron domains than single bonds.
- They exert a greater 施加 repulsive force than single bonds, making their bond angles greater.

Expanding beyond the Octet Rule

- Remember that some elements can break the octet rule and make *more* than four bonds (or have more than four electron domains).
- The result is two more possible electron domains: five = trigonal bipyramidal;
six = octahedral (as was seen in the slide on electron-domain geometries).

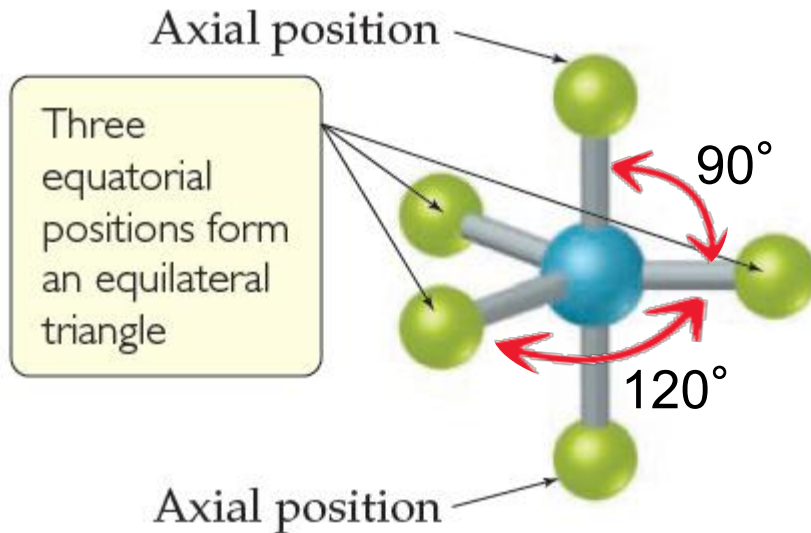
三角双锥体

八面体

电子畴几何形状

Trigonal Bipyramidal Electron Domain

三角双锥电子域

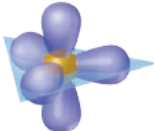
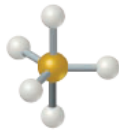
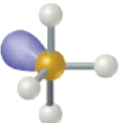


- There are two distinct positions in this geometry:
 - Axial 轴向
 - Equatorial 赤道
- Lone pairs occupy equatorial positions.
(an equatorial domain experiences less repulsion than an axial domain)

Trigonal Bipyramidal Electron Domain

- There are four distinct molecular geometries in this domain:
 - Trigonal bipyramidal
 - Seesaw
 - T-shaped
 - Linear

Table 9.3 Electron-Domain and Molecular Geometries for Five and Six Electron Domains around a Central Atom

Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
5	 Trigonal bipyramidal	5	0	 Trigonal bipyramidal	PCl ₅
		4	1	 Seesaw	SF ₄
		3	2	 T-shaped	ClF ₃
		2	3	 Linear	XeF ₂

Seesaw（跷跷板形）在化学中是一种分子几何构型，也称为扭曲四面体或不规则四面体。

Octahedral Electron Domain

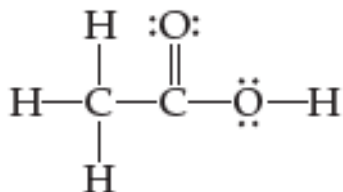
Table 9.3 Electron-Domain and Molecular Geometries for Five and Six Electron Domains around a Central Atom

Number of Electron Domains	Electron-Domain Geometry	Bonding Domains	Nonbonding Domains	Molecular Geometry	Example
6		6	0		SF_6
	Octahedral			Octahedral	
		5	1		BrF_5
				Square pyramidal	
		4	2		XeF_4
				Square planar	

- All positions are equivalent in the octahedral domain.
- There are three molecular geometries:
 - Octahedral 八面体
 - Square pyramidal 方形金字塔
 - Square planar 平面正方形

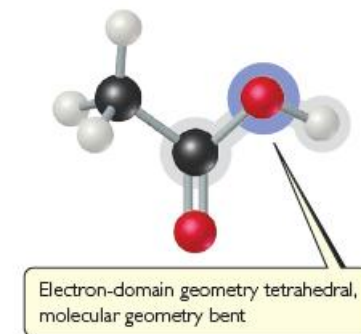
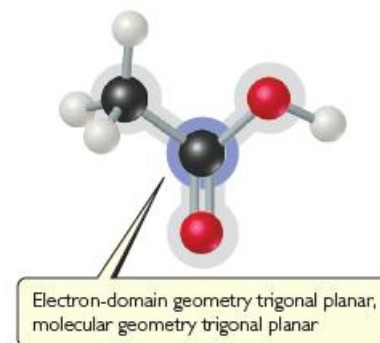
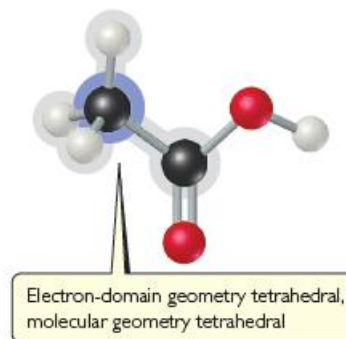
Shapes of Larger Molecules

For larger molecules, **look at the geometry about each atom** rather than the molecule as a whole.



	$ \begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}- \\ \\ \text{H} \end{array} $	$ \begin{array}{c} \text{:O:} \\ \\ \text{C}- \end{array} $	$ \begin{array}{c} \ddot{\text{O}}-\text{H} \end{array} $
Number of electron domains	4	3	4
Electron-domain geometry	Tetrahedral 四面体	Trigonal planar	Tetrahedral
Predicted bond angles	109.5°	120°	109.5°

平面三角形



Polarity of Molecules

Ask yourself:

COVALENT or IONIC? If COVALENT:

Are the BONDS polar?

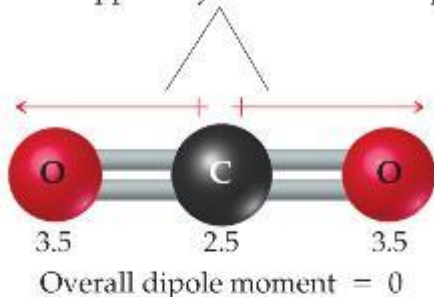
- a. NO: The molecule is NONPOLAR!
- b. YES: Continue—Do the AVERAGE position of $\delta+$ and $\delta-$ coincide? 同时发生
 - 1) YES: The molecule is NONPOLAR.
 - 2) NO: The molecule is POLAR.

NOTE: Different atoms attached to the central atom have different polarity of bonds.

Comparison of the Polarity of Two Molecules

A NONPOLAR molecule

Equal and oppositely directed bond dipoles



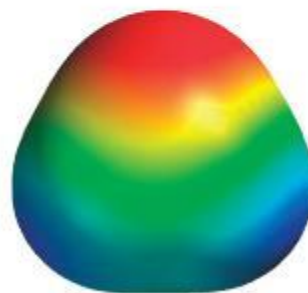
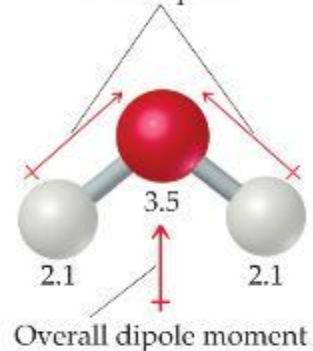
Low electron density

High electron density



A POLAR molecule

Bond dipoles

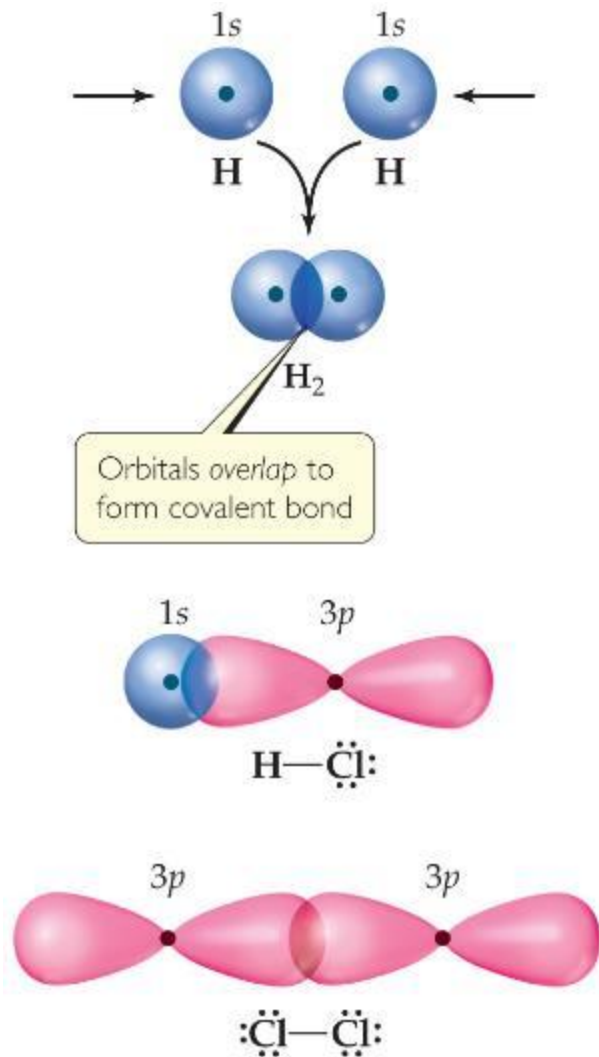


Low electron density

High electron density



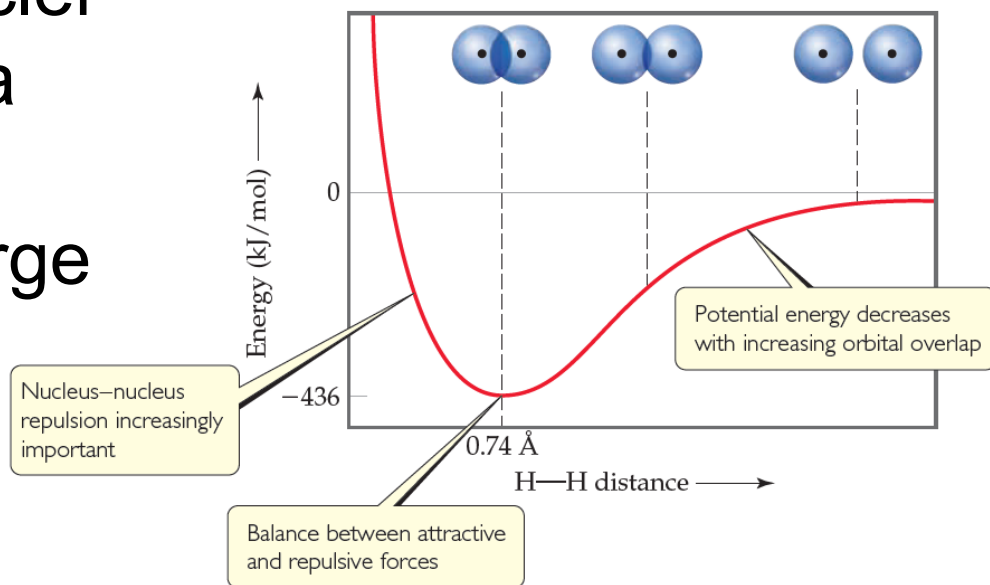
Valence-Bond Theory



- In Valence-Bond Theory, electrons of two atoms begin to occupy the same space.
- This is called “overlap” of orbitals.
- The sharing of space between two electrons of opposite spin results in a covalent bond.

Overlap and Bonding

- Increased overlap brings the electrons and nuclei closer together until a balance is reached between the like charge repulsions and the electron-nucleus attraction.
- Atoms can't get too close because the internuclear repulsions get too great.



VSEPR and Hybrid Orbitals

- VSEPR predicts shapes of molecules very well.
- How does that fit with orbitals? 轨道
- Let's use H₂O as an example:
- If we draw the best Lewis structure to assign VSEPR, it becomes bent. 结构
- If we look at oxygen, its electron configuration is $1s^2 2s^2 2p^4$. If it shares two electrons to fill its valence shell, they should be in $2p$. 价层电子
- Wouldn't that make the angle 90° ?
- Why is it 104.5° ?

Hybrid Orbitals 杂化轨道

- To **explain molecular geometries**, we **assume** that the atomic orbitals on an atom mix to form new orbitals called **hybrid orbitals**.
- **Hybrid orbitals** form by “mixing” of atomic orbitals to create new orbitals of equal energy, called degenerate orbitals. 简并轨道
- When two orbitals “mix”, they create two orbitals; when three orbitals “mix”, they create three orbitals; etc.

Be—*sp* hybridization

- When we look at the orbital diagram for beryllium (Be), we see that there are only paired electrons in full sub-levels.
- Be makes electron deficient compounds with two bonds for Be. Why? *sp* hybridization (mixing of one *s* orbital and one *p* orbital)

the orbital diagram for a ground-state Be atom



1s

2s

2p

envision to “promote” one 2s e- to a 2p orbital



1s

2s

2p

hybridize one *s* orbital and one *p* orbital to form two degenerate *sp* hybrid orbitals



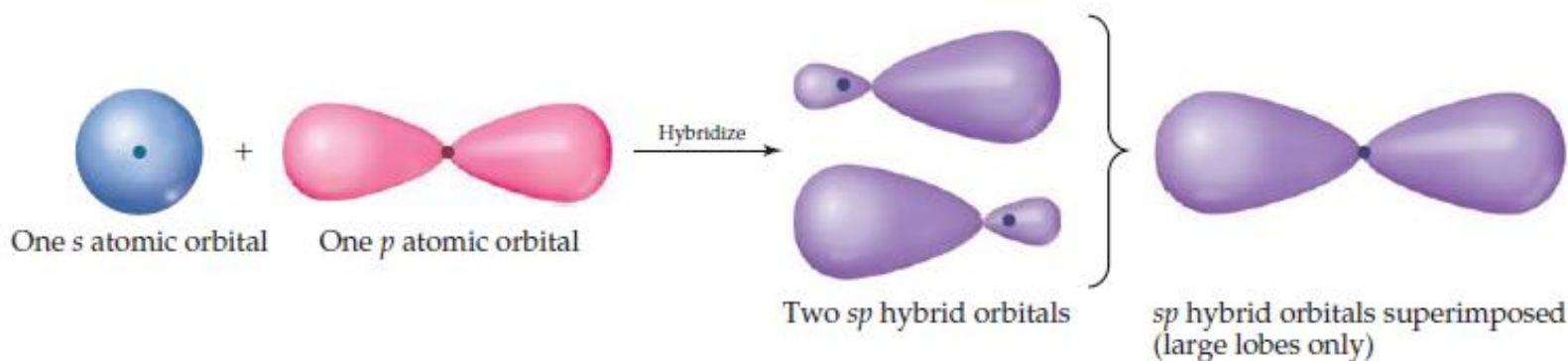
1s

sp

2p

sp Orbitals

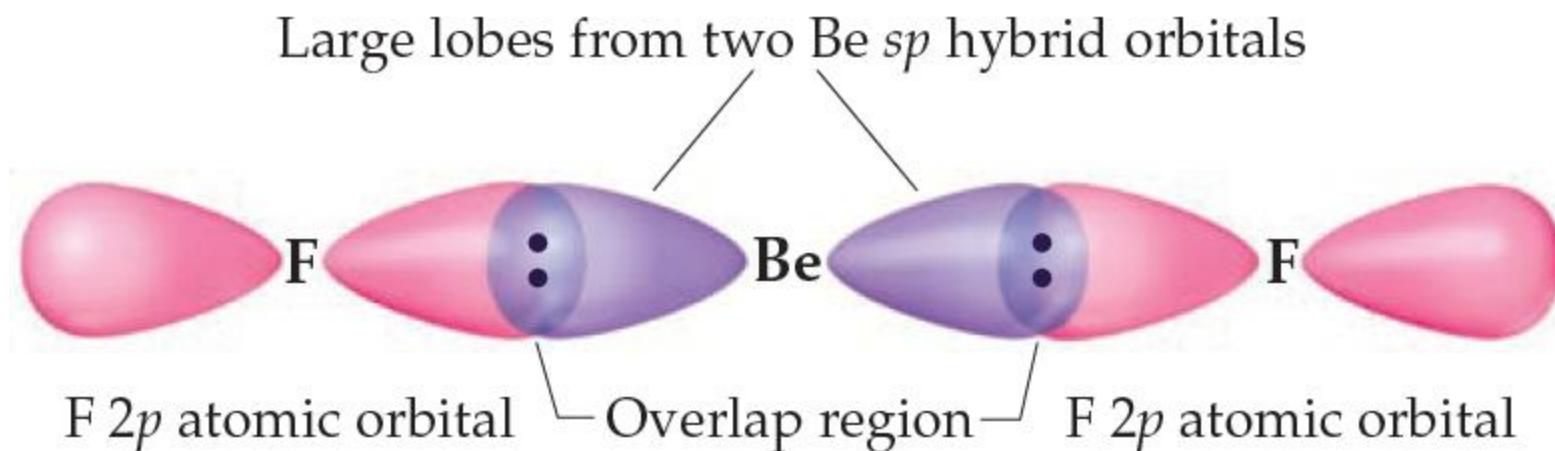
- Mixing the s and p orbitals yields two degenerate orbitals that are hybrids of the two orbitals.
 - These sp hybrid orbitals have two lobes like a p orbital. 叶瓣
 - One of the lobes is larger and more rounded, as is the s orbital.



Position of sp Orbitals

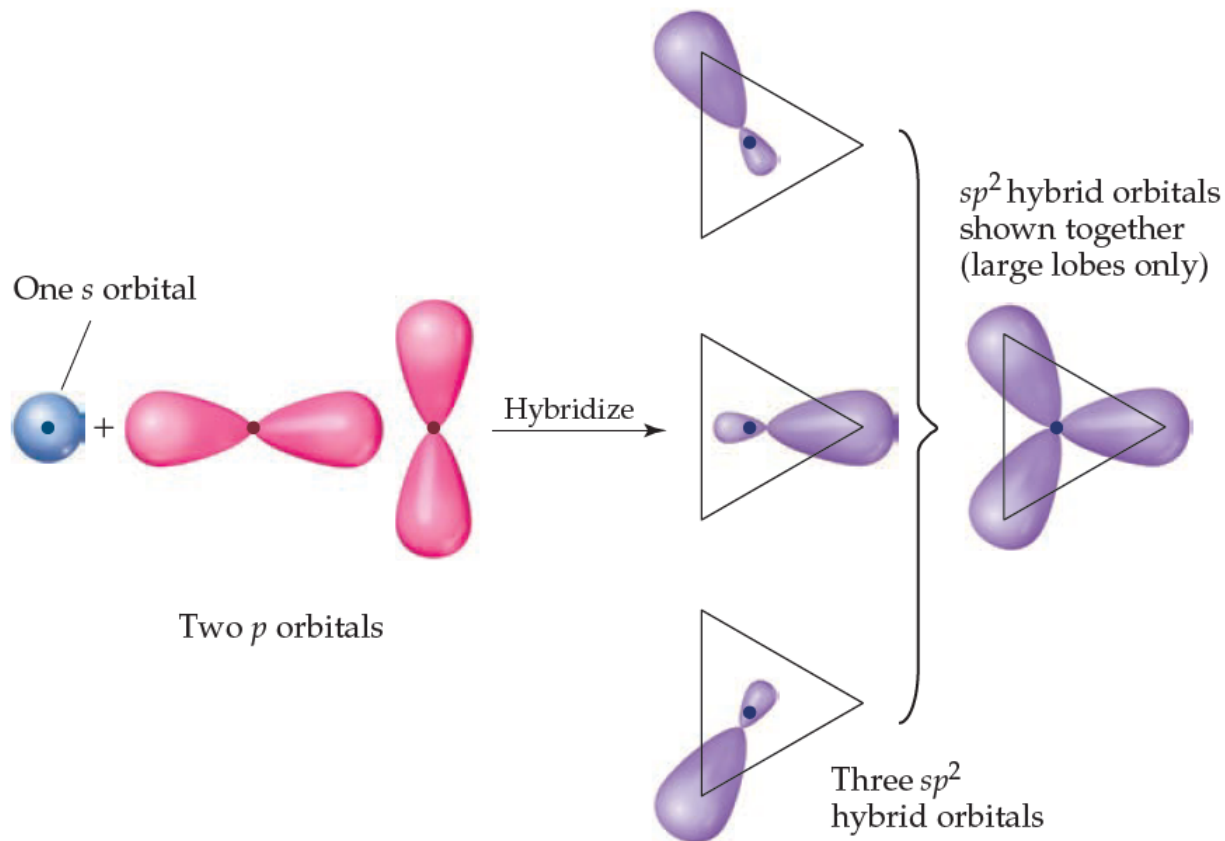
使—形成一条线

- These two degenerate orbitals would align themselves 180° from each other.
- This is consistent with the observed geometry of Be compounds (like BeF_2) and VSEPR: linear.



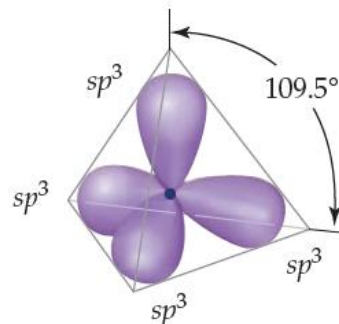
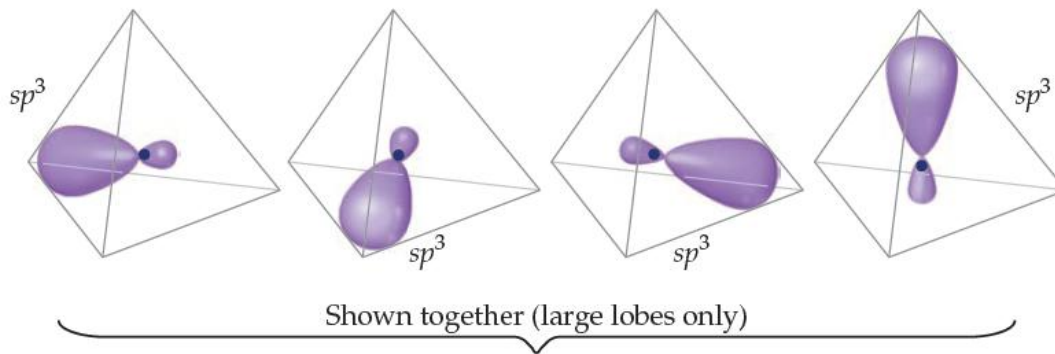
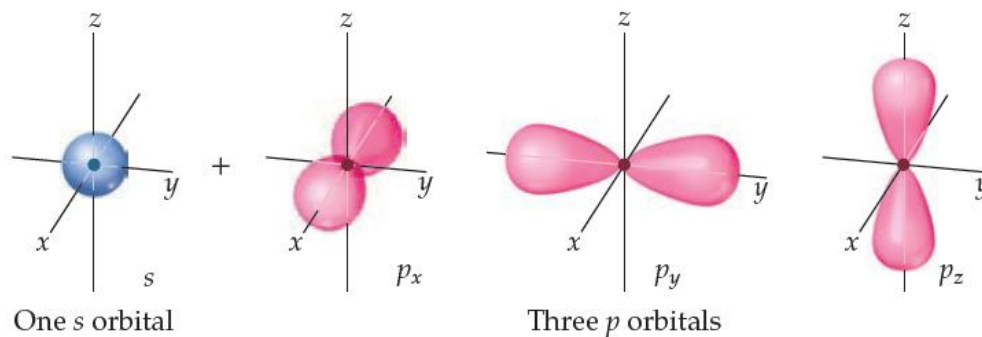
Boron—Three Electron Domains Gives sp^2 Hybridization

Using a similar model for boron leads to three degenerate sp^2 orbitals.



Carbon: sp^3 Hybridization

With carbon,
we get four
degenerate
 sp^3 orbitals.



Hypervalent Molecules 超价分子

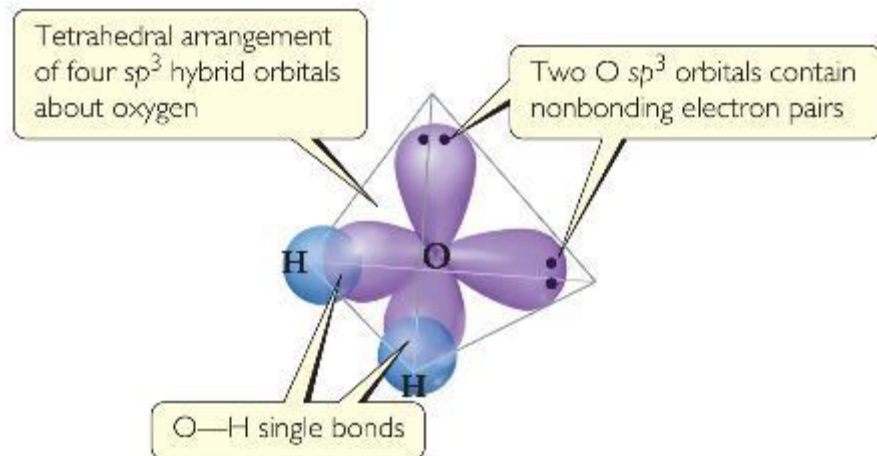
- The elements which have *more* than an octet
- Valence-Bond model would use *d* orbitals to make more than four bonds.
- This view works for period 3 and below.
- Theoretical studies suggest that the energy needed would be too great for this.
- A more detailed bonding view is needed than we will use in this course.

“This discussion reminds us that **models in science are not reality** but rather are our attempts to describe aspects of reality that we have been able to measure, such as bond distances, bond energies, molecular geometries, and so on. A model may work well up to a certain point but not beyond it, as is the case for hybrid orbitals. The hybrid orbital model for period 2 elements has proven very useful and is an essential part of any modern discussion of bonding and molecular geometry in **organic chemistry**. When it comes to molecules such as SF_6 , however, we encounter the **limitations** of the model.”

--- quote from “Chemistry, the Central Science”

What Happens with Water?

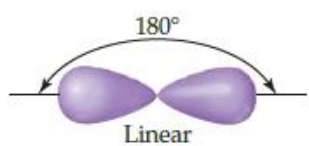
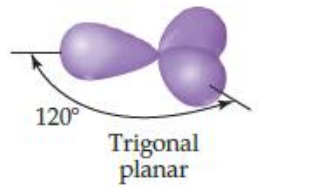
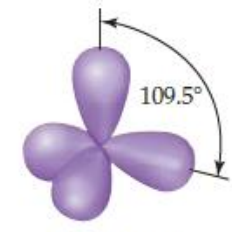
- We started this discussion with H_2O and the angle question: Why is it 104.5° instead of 90° ?
- Oxygen has two bonds and two lone pairs—four electron domains.
- The result is sp^3 hybridization!



Hybrid Orbital Summary

- 1) Draw the Lewis structure.
- 2) Use VSEPR to determine the electron-domain geometry.
- 3) Specify the hybrid orbitals needed to accommodate these electron pairs based on their geometric arrangement.

Table 9.4 Geometric Arrangements Characteristic of Hybrid Orbital Sets

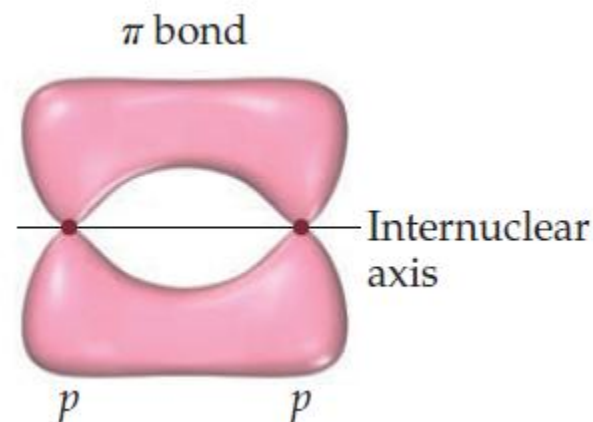
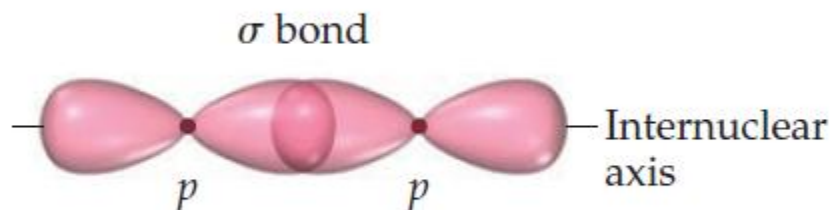
Atomic Orbital Set	Hybrid Orbital Set	Geometry	Examples
s, p	Two sp	 Linear	$\text{BeF}_2, \text{HgCl}_2$
s, p, p	Three sp^2	 Trigonal planar	BF_3, SO_3
s, p, p, p	Four sp^3	 Tetrahedral	$\text{CH}_4, \text{NH}_3, \text{H}_2\text{O}, \text{NH}_4^+$

2 electron domains sp hybridization
 3 electron domains sp^2 hybridization
 4 electron domains sp^3 hybridization

Types of Bonds

- How does a double or triple bond form?
- It *can't*, if we only use hybridized orbitals.
- *However*, if we use the orbitals which are *not* hybridized, we can have a “side-ways” overlap.
- Two types of bonds:
 - Sigma (σ) bond
 - Pi (π) bond

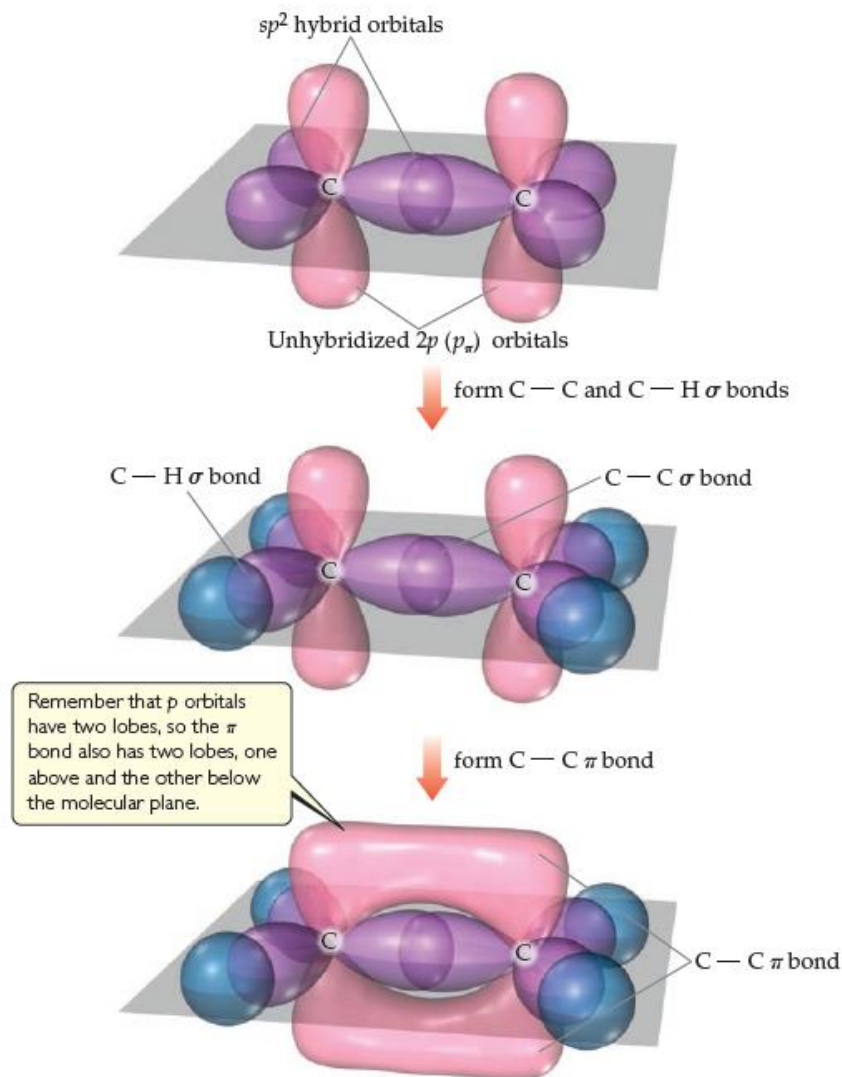
Sigma (σ) and Pi (π) Bonds



- Sigma bonds are characterized by
 - head-to-head overlap.
 - 圆柱形的 对称性
cylindrical symmetry of electron density about the internuclear axis. 电子密度围绕核间轴呈圆柱形对称
- Pi bonds are characterized by
 - 核间的 轴、轴线
side-to-side overlap. 轨道重叠
 - electron density above and below the internuclear axis.

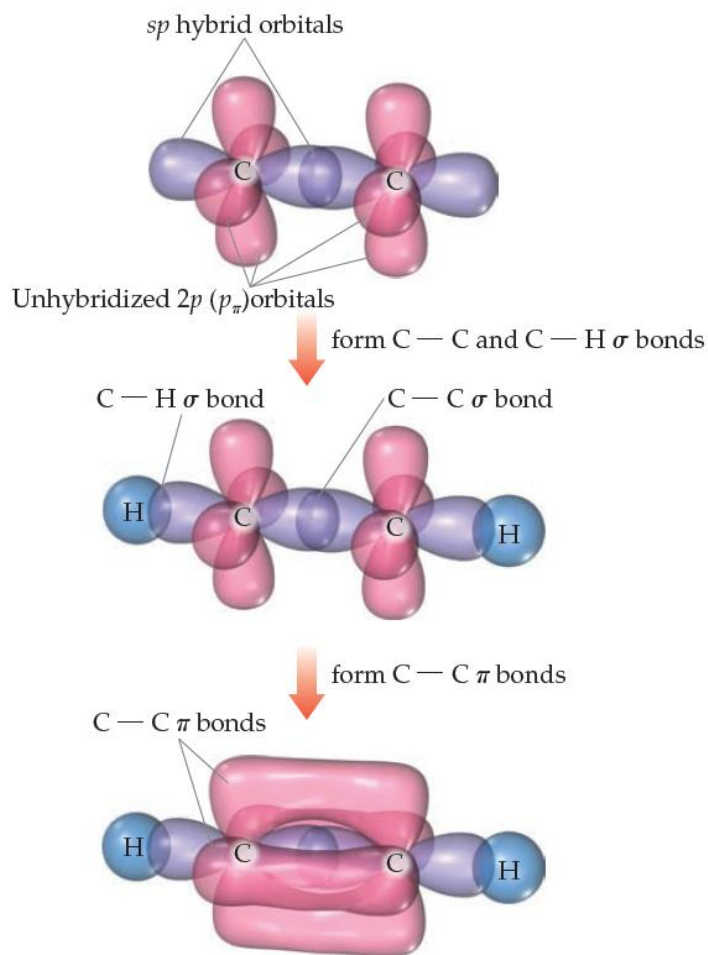
Bonding in Molecules

- Single bonds are always σ -bonds.
- Double bonds have one σ -bond, one π -bond.



▲ Figure 9.23 The orbital structure of ethylene, C_2H_4 .

Bonding in Molecules



▲ Figure 9.24 Formation of two π bonds in acetylene, C_2H_2 .

- Single bonds are always σ -bonds.
- Double bonds have one σ -bond, one π -bond.
- Triple bonds have one σ -bond, two π -bonds.

Localized or Delocalized Electrons

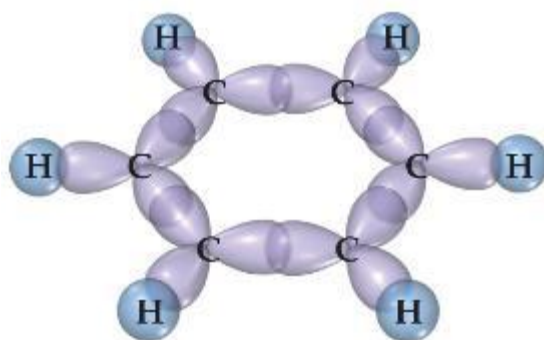
局域电子

离域电子

- Bonding electrons (σ or π) that are specifically shared between two atoms are called **localized** electrons.
- In many molecules, we can't describe all electrons that way (resonance); the other electrons (shared by multiple atoms) are called **delocalized** electrons.

Benzene

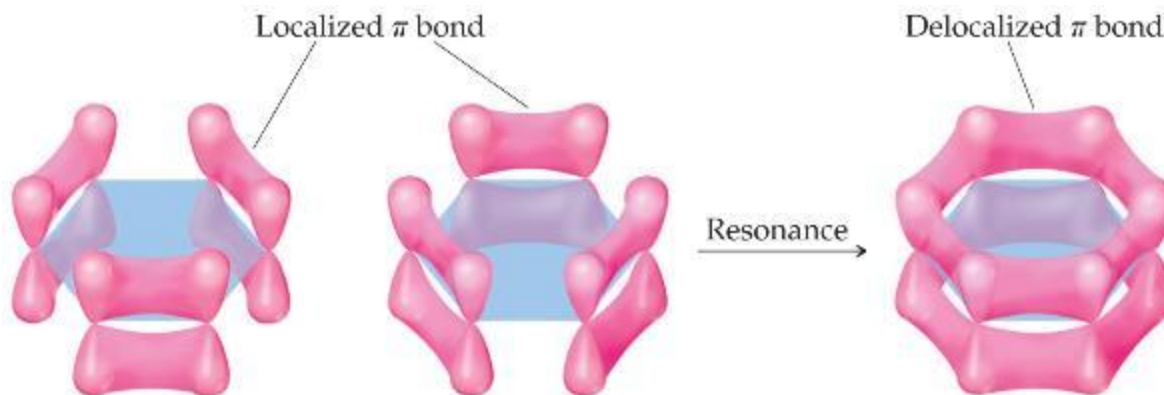
The organic molecule benzene (C_6H_6) has six σ -bonds and a p orbital on each C atom, which form delocalized bonds using one electron from each p orbital.



(a) σ bonds



(b) p_{π} orbitals



Molecular Orbital (MO) Theory

分子轨道理论

波的性质

- Wave properties are used to describe the energy of the electrons in a molecule.
- **Molecular orbitals** have many characteristics like atomic orbitals:
 - maximum of two electrons per orbital
 - Electrons in the same orbital have opposite spin.
 - Definite energy of orbital
 - Can visualize electron density by a contour diagram

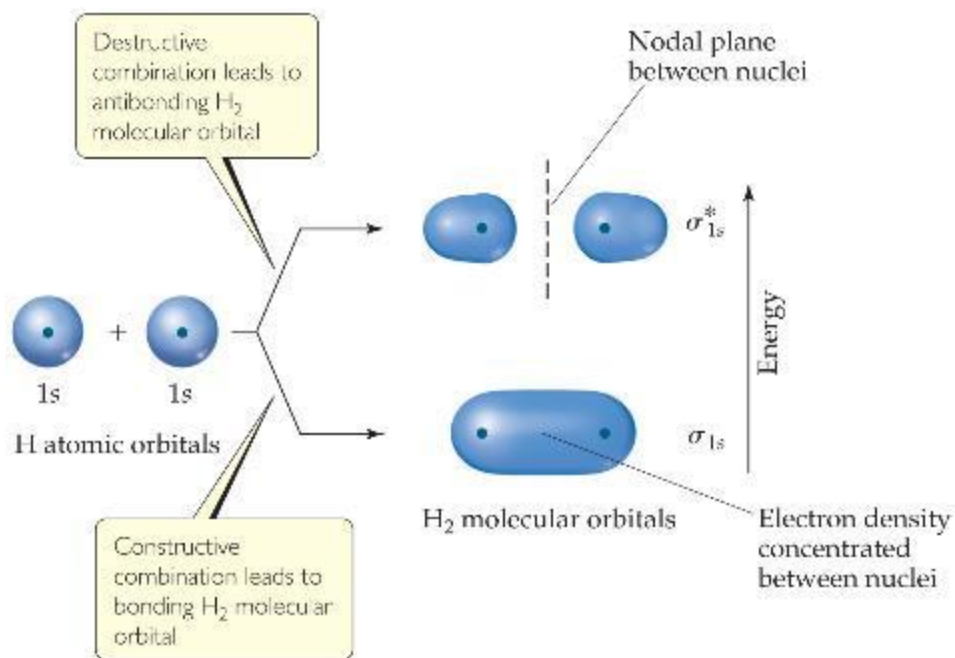
More on MO Theory

- They differ from atomic orbitals because they represent the entire molecule, not a single atom.
- Whenever two atomic orbitals overlap, two molecular orbitals are formed: one bonding, one antibonding.
- **Bonding orbitals** ^{成键轨道} are constructive combinations of atomic orbitals.
- **Antibonding orbitals** ^{反键轨道} are ^{破坏性的} destructive combinations of atomic orbitals. They have a new feature unseen before: A **nodal plane** occurs where electron density equals zero.

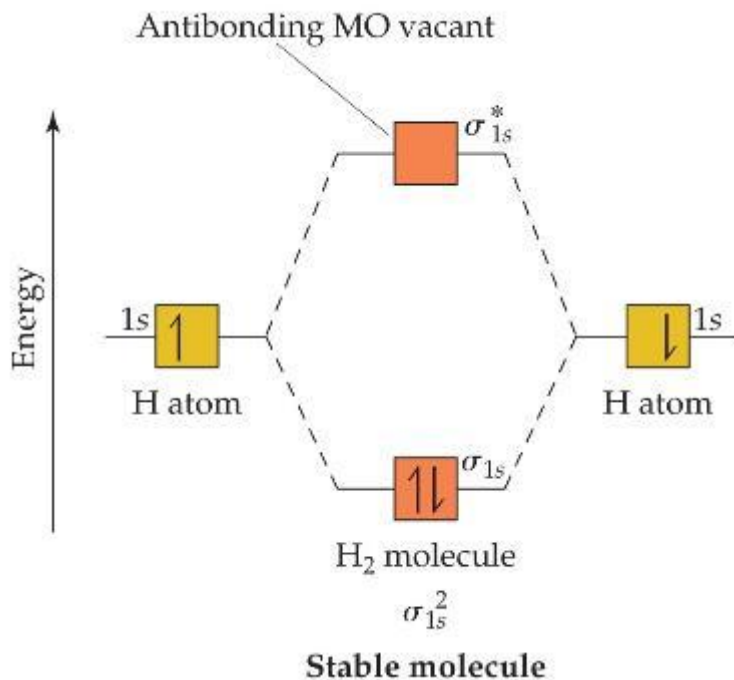
反键轨道是原子轨道的相消组合。它们具有一个前所未有的新特征：出现了一个电子密度等于零的节面

Molecular Orbital (MO) Theory

Whenever there is direct overlap of orbitals, forming a bonding and an antibonding orbital, they are called **sigma (σ) molecular orbitals**. The antibonding orbital is distinguished with an asterisk as σ^* . Here is an example for the formation of a hydrogen molecule from two atoms.



MO Diagram



- An **energy-level diagram**, or **MO diagram** shows how orbitals from atoms combine to give the molecule.
- In H₂ the two electrons go into the bonding molecular orbital (lower in energy).
- **Bond order** = $\frac{1}{2}(\text{\# of bonding electrons} - \text{\# of antibonding electrons}) = \frac{1}{2}(2 - 0) = 1 \text{ bond}$

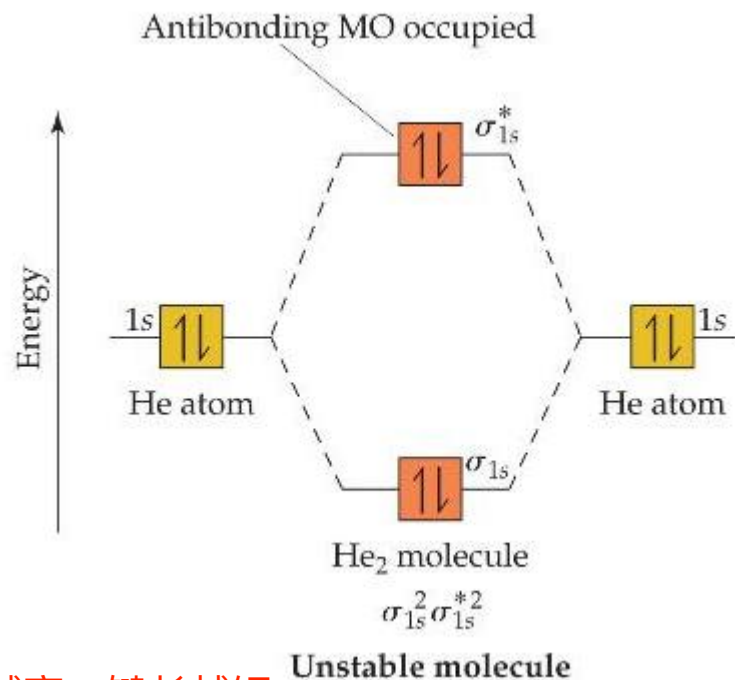
键级 (Bond Order) 是衡量分子中化学键强度的一个重要参数，用于描述两个原子之间成键的牢固程度。

核心定义

键级 = $\frac{1}{2} \times (\text{成键轨道中的电子数} - \text{反键轨道中的电子数})$

Can He₂ Form? Use MO Diagram and Bond Order to Decide!

- Bond Order = $\frac{1}{2}(2 - 2)$
= 0 bonds
- Therefore, He₂ does *not* exist.



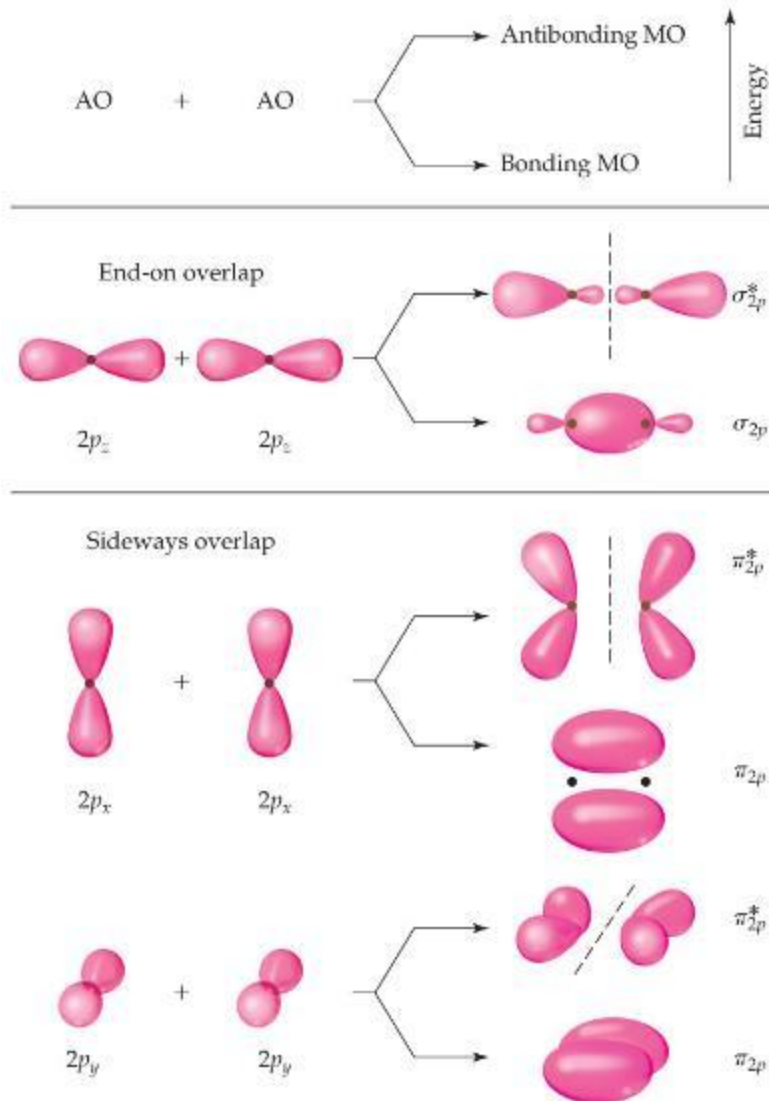
关键意义

键级越大，化学键越强、键能越高、键长越短

键级可以是整数（如单键 = 1，双键 = 2），也可以是分数（如某些共轭体系或自由基）

键级为 0 意味着无法形成稳定分子（如 He₂）

s and p Orbitals Can Interact

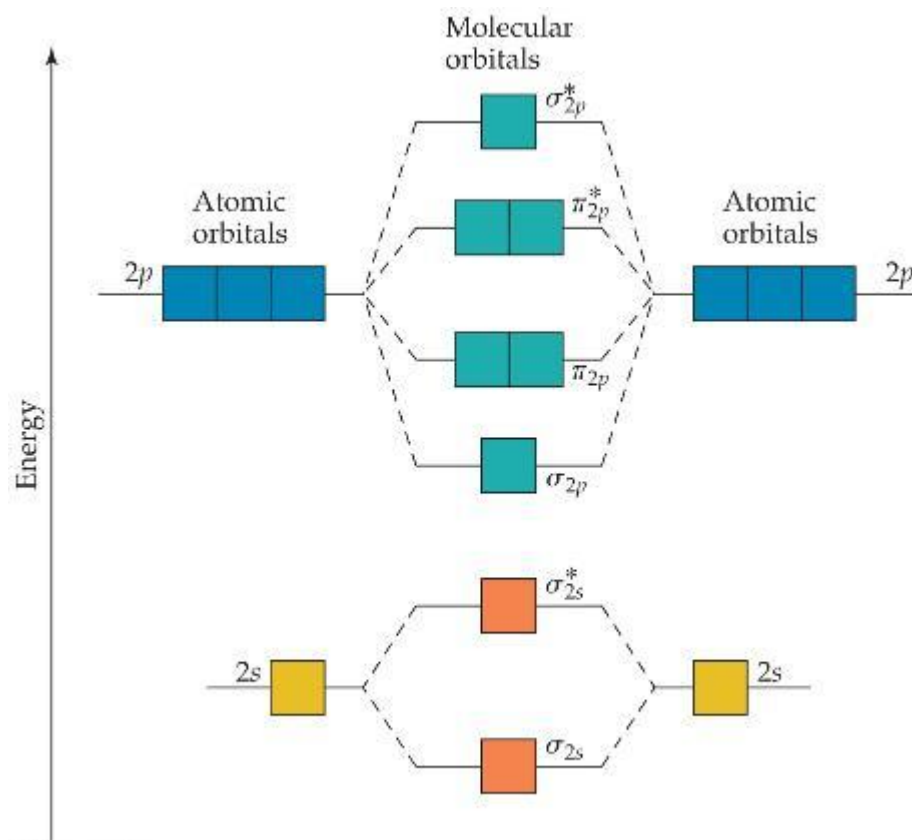


- For atoms with both s and p orbitals, there are two types of interactions:
- The s and the p orbitals that face each other overlap in σ fashion.
- The other two sets of p orbitals overlap in π fashion.
- These are, again, direct and “side-ways” overlap of orbitals.

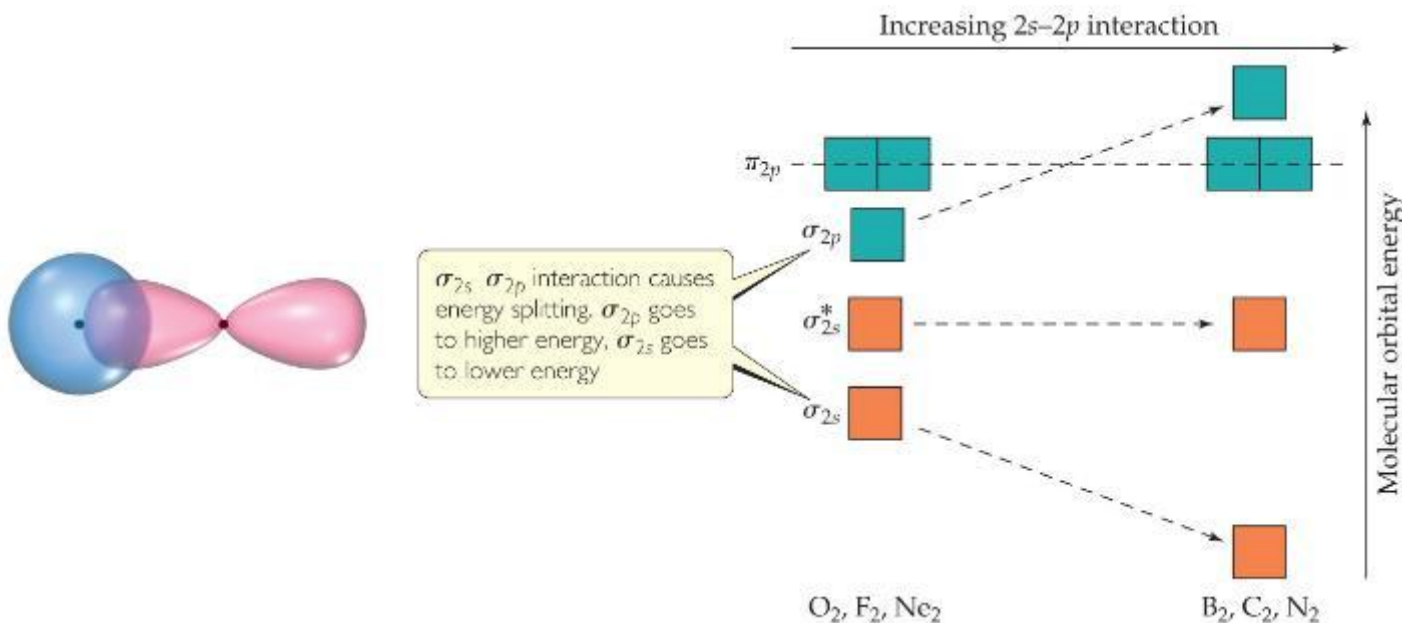
MO Theory

- The resulting MO diagram:

- There are σ and σ^* orbitals from s and p atomic orbitals.
- There are π and π^* orbitals from p atomic orbitals.
- Since direct overlap is stronger, the effect of raising and lowering energy is greater for σ and σ^* .

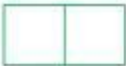
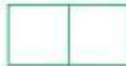
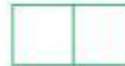
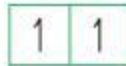
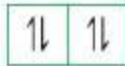
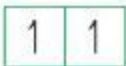
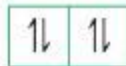
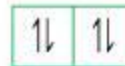


s and p Orbital Interactions



- In some cases, s orbitals can interact with the p_z orbitals more than the p_x and p_y orbitals.
- It raises the energy of the p_z orbital and lowers the energy of the s orbital.
- The p_x and p_y orbitals are degenerate orbitals.

MO Diagrams for Diatomic Molecules of 2nd Period Elements

	Large 2s–2p interaction			Small 2s–2p interaction		
	B ₂	C ₂	N ₂	O ₂	F ₂	Ne ₂
σ_{2p}^*						
π_{2p}^*						
σ_{2p}						
π_{2p}						
σ_{2s}^*						
σ_{2s}						
Bond order	1	2	3	2	1	0
Bond enthalpy (kJ/mol)	290	620	941	495	155	—
Bond length (Å)	1.59	1.31	1.10	1.21	1.43	—
Magnetic behavior	Paramagnetic	Diamagnetic	Diamagnetic	Paramagnetic	Diamagnetic	—

MO Diagrams and Magnetism

抗磁性是由于每个轨道中的所有电子都自旋配对的结果。这些物质会被磁场微弱地排斥

- **Diamagnetism** is the result of all electrons in every orbital being spin paired. These substances are weakly repelled by a magnetic field.

抗磁性

向。 排列，使物质被磁场微弱吸引

顺磁性是由于轨道中存在一个或多个未成对电子的结果，这些未成对电子的磁矩会在外磁场中沿磁场方

- **Paramagnetism** is the result of the presence of one or more unpaired electrons in an orbital.

顺磁性

顺磁性：有未成对电子，磁矩沿外磁场方向排列，产生吸引

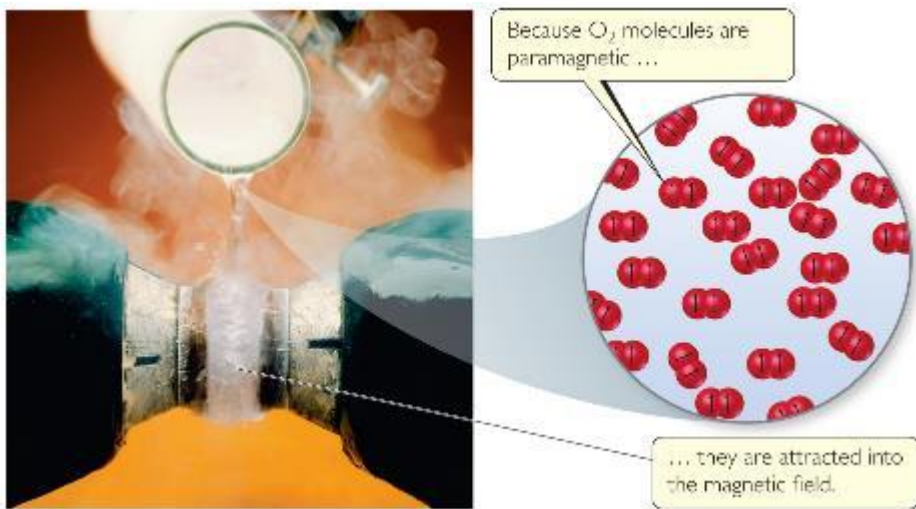
抗磁性：无未成对电子，仅产生微弱排斥

铁磁性：原子磁矩自发有序排列，产生强吸引力

- Is oxygen (O_2) paramagnetic or diamagnetic? Look back at the MO diagram! It is paramagnetic.

Paramagnetism of Oxygen

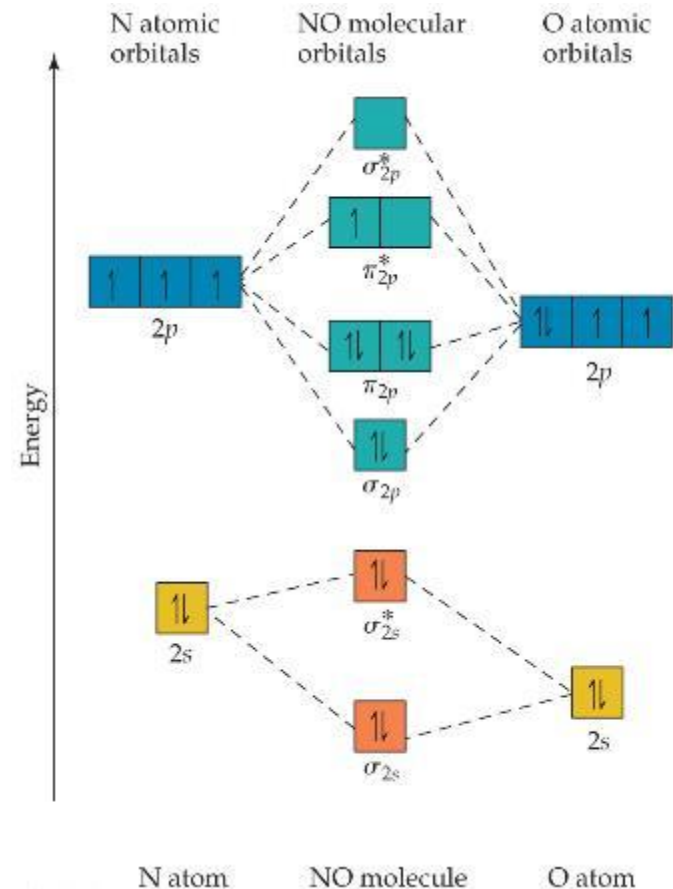
- Lewis structures would *not* predict that O_2 is paramagnetic.
- The MO diagram clearly shows that O_2 is paramagnetic.
- Both show a double bond (bond order = 2).



Heteronuclear Diatomic Molecules

异核双原子分子

- Diatomic molecules can consist of atoms from different elements.
- How does a MO diagram reflect differences?
- The atomic orbitals have different energy, so the interactions change slightly.
- The more electronegative atom has orbitals lower in energy, so the bonding orbitals will more resemble them in energy.



异核双原子分子是由两种不同元素的原子组成的双原子分子，如 HF、HCl、CO、NO 等

How to Determine the Hybridization

